Problem 18.6

A sum P is used to buy a deferred perpetuity-due of 1 payable annually. The annual effective interest rate is $i > 0$. Find an expression for the deferred period.

Solution

Let m be the deferred period. A perpetuity-due of 1 payable annually has value

$$\ddot{a}_{\overline{\infty}|} = \frac{1}{d},$$

where

$$d = \frac{i}{1+i}.$$

If the perpetuity-due is deferred for m years, its present value is

$$P = v^m \ddot{a}_{\overline{\infty}|}.$$

Thus,

$$P = v^m \left(\frac{1}{d}\right).$$

Multiplying both sides by d gives

$$Pd = v^m.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$Pd = (1+i)^{-m}.$$

Taking natural logarithms,

$$\ln(Pd) = -m\ln(1+i).$$

Therefore,

$$m = \frac{-\ln(Pd)}{\ln(1+i)}.$$

Substituting $d = \dfrac{i}{1+i}$ gives

$$m = \frac{-\ln\left(P\dfrac{i}{1+i}\right)}{\ln(1+i)}.$$

Hence, the deferred period is

$$\boxed{m = \frac{-\ln\left(P\dfrac{i}{1+i}\right)}{\ln(1+i)}}.$$